CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

April/May 2013

EC304 Power Electronics

Date: 18.04.2013, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

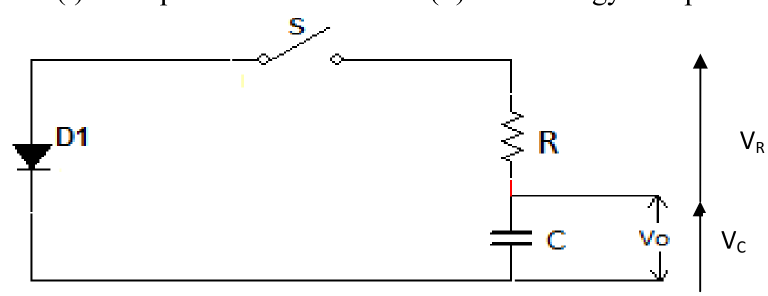
Q - 1

(a) Explain the operation of SCR using two transistor analogy by deriving required equations. **[04]**

(b) Justify the statement ***"In three phase half wave control rectifier the required minimum firing angle is α=30"***. **[03]**

Q - 2 (a) A diode circuit shown in fig. given below with R=44Ω and C=0.1µF. The capacitor has an initial voltage, Vo=220v. If switch S_1 is closed at t=0, determine **[04]**

(i) the peak diode current (ii) the energy dissipated in the resistor R



(b) Enumerate the methods by which SCR can be turned on. **[04]**

(c) Explain 1-Ø bidirectional ac voltage controller with resistive load. Draw necessary diagram and derive output voltage equation. **[06]**

OR

Q - 2 (a) Why heat is generated in within the power devices? Explain electrical anlog of heat transfer from a power semiconductor device. **[04]**

(b) Explain working principle, application of following thyristor with symbol **[04]**

(a) GTO (b) LASCR

(c) What is cycloconverter ? Discuss its operation principle and sketch the diagram for 1-Ø to 1-Ø cycloconvertor. **[06]**

Q - 3 (a) What are the types of power diodes? Explain any one in detail. **[03]**

(b) Explain how the isolation is accomplished between the thyristor and its gate-pulse-generator circuit. **[03]**

(c) Draw and explain 3-Ø half wave control rectifier circuit. Draw its waveform for following firing angle **[08]**
(a) $\alpha=15^0$ (b) $\alpha=45^0$

OR

Q - 3 (a) List the advantages of COOLMOS over conventional MOSFET. **[03]**

(b) Discuss the following with respect to BJT **[03]**
i) Secondary breakdown ii) Safe operating area

(c) Explain single phase semi converter for highly inductive load with the help of circuit diagram. Draw supply voltage, output voltage waveforms. Write equation for average output voltage and RMS value of output voltage. **[08]**

SECTION – II

Q - 4(a) List the control strategies for operating the switches employed in DC choppers. Explain any one detail. **[03]**

(b) Explain the operating modes of DC drives. **[04]**

Q - 5 (a) Explain the principle of working of a single phase half bridge inverter with the help of the circuit and waveforms for both resistive and inductive loads. Calculate rms output voltage at the fundamental frequency and the output power for a dc input voltage of 48 V and resistive load of 2.4 Ω. **[08]**

(b) Classify the chopper. List all types of Chopper and explain any one of them. **[06]**

OR

Q - 5 (a) Draw and explain the structure of power MOSFET. Also explain principal of operation and switching characteristics of power MOSFET. **[08]**

(b) Draw and explain the block diagram for microcomputer control of DC drives **[06]**

Q - 6 (a) Draw the circuit of boost regulator and explain its working with equivalent circuit and waveforms. Obtain equation for ΔI and ΔVc. **[08]**

(b) Explain Single phase Full (Bridge) controlled rectifier with resistive load in detail. **[06]**

OR

Q - 6 (a) Describe the operation of 3-phase inverter with 120^0 conduction. **[08]**

(b) What is the principle of operation of step-down converter? Show complete analysis of step-down converter with RL load with all necessary diagrams. **[06]**

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